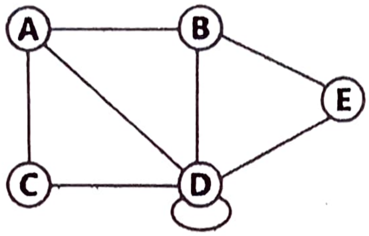
Reference No. Ex/IT/PC/B/T/224/2023

Bachelor Of Engineering In Information Technology
2nd Year 2nd Semester, Semester Examination, 2023
Subject Name –Graph Theory & Combinatorics (IT/PC/B/T/224)

Full Marks=100

CO1 [10]

Q1. (i) Find the adjacency matrix of the graph given below. Find out the number of paths of length 2 from one vertex to others.



(ii) Find out the maximum and minimum number of edges in a 3-component graph having 10 vertices.

[5+5=10]

CO2 [20]

Q2. (i) Does a Hamiltonian path or circuit exist on the graph below (Figure: 1)? Justify it.

(ii) Prove with a suitable example "Every tree has either one or two centers".

(iii) Using Prim's algorithm find the minimal spanning tree of the graph given below (Figure: 2).

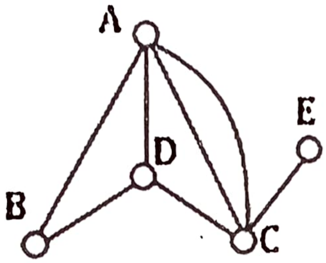


Figure: 1

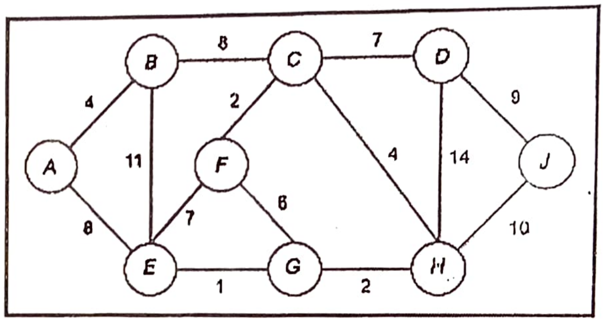


Figure: 2

The vertex connectivity of a graph G refers to the minimum number of vertices that need to be removed in order to disconnect the graph. The edge connectivity of a graph G, on the other hand, refers to the minimum number of edges that need to be removed in order to disconnect the graph.

[6+6+8=20]

CO3 [20]

Q3. (i) Does a cut point and cut edges exist of the graph given below (Figure: 3)? If yes then mention them.

(ii) Find the union of the two graphs G_1 and G_2 given below (Figure: 4).

(iii) Justify this using a suitable example "The vertex connectivity of any graph G can never exceed the edge connectivity of G".

Example - Consider a complete graph with 4 vertices (denoted as K4). The vertex connectivity of K4 is 3, as removing any 3 vertices will disconnect the graph. The edge connectivity of K4 is also 3, as removing any 3 edges will disconnect the graph. In this example, the vertex connectivity is equal to the edge connectivity.

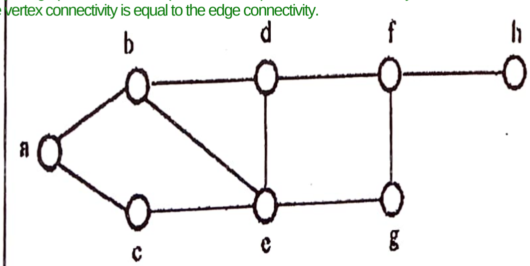


Figure: 3

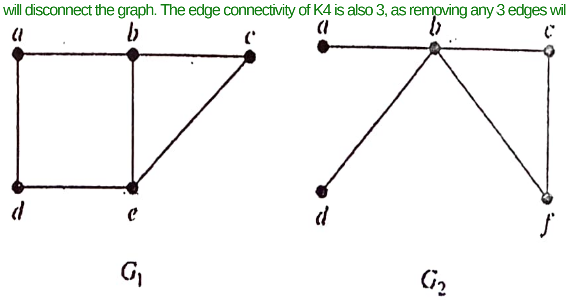


Figure: 4

[7+7+6=20]

CO4 [20]

Q4. (i) "A graph is 2-colorable if it is bipartite and every cycle has an even length." Justify this with a suitable example.

(ii) If a graph has 100 vertices and 300 edges, can it be planer?

(iii) Find the chromatic number of the graph given below (Figure: 5). (Mention all steps properly)

[Turn over

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(iv) Find out how many vertices can be matched using maximum matching in the bipartite graph algorithm of the following graph given below (Figure: 6)? Can we call it perfect matching? Justify this.

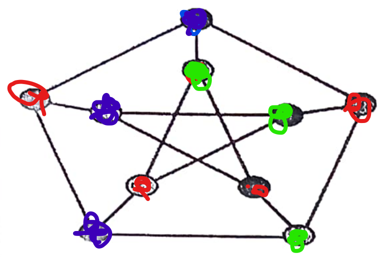

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Figure: 5

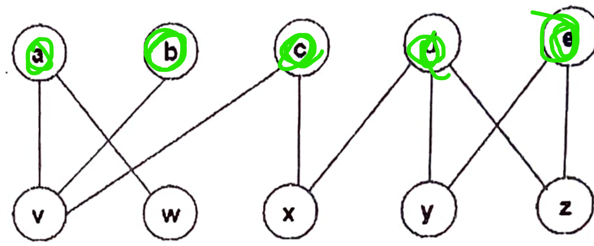

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Figure: 6

[6+3+4+(4+3)=20]

CO5 [20]

Q5. (i) How many cards must be selected from a standard deck of 52 cards to guarantee that at least three hearts are selected?? 8*2+1

(ii) How many ways are there for **eight men** and **five women** to stand in a line so that **no two women** stand next to each other? 8! x 9c5 x 5!

(iii) Suppose a club has 25 members. How many ways are there to choose a **president, vice-president, secretary**, and **treasurer** of the club, where no person can hold more than one office?

25 x 24 x 23 x 22

[6+7+7=20]

CO6 [10]

Q6.(i) Find the generating functions of the following sequences given below in closed form .

$$0,\ 1,\ -\frac{1}{2},\ \frac{1}{3},\ -\frac{1}{4},\ \ldots$$

(ii) Using generating functions, find a_n in terms of n for the case given below:

$$a_0 = 1,\ a_1 = 2 \text{ and } a_{n+2} = 5a_{n+1} - 4a_n \text{ for } n \ge 0$$

[5+5=10]

CO1: **Explain** and **discuss** the concept of different types of Graphs with fundamental properties and **express** different types of matrix representation. (K2)

CO2: Illustrate different types of trees such as (i) rooted tree (ii) spanning tree etc, and **explain** their properties. (K3)

CO3: Apply operations like Union, Deletion, and decomposition of graphs and **illustrate** Cut vertex and Cut edge and their properties. (K3)

CO4: Illustrate planar graph and their properties and Graph Coloring and Matching. (K3)

CO5: Apply and **evaluate** basic counting rules, pigeon-hole principle and principle of inclusion-exclusion. (K3)

CO6: Apply and Solve problems using Generating Function and Recurrence Relations. (K3)